Heaven's Light is Our Guide

Rajshahi University of Engineering and Technology



Course Code
ECE 3112

Course Title
Microprocessor, Assembly Language & Interfacing Sessional

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Lab Report 6: Using Assembly Language to Program 8086 Microprocessor to Solve Problems Related To Shifting & Stack.

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Problem 6.1

Problem Description

Write a code to take two input from user as Binary and multiply them using shifting operation and output the result in binary.

Tools Used

- emu8086 - 8086 Microprocessor Emulator
- MacTeX -LaTeXcompiler.
- VS Code with LaTeXworkshop extension as a text editor.

Process

Code:

```
—

DATA SEGMENT
    INA DW ?
    INB DW ?
    COUNT DB 16
DATA ENDS

CODE SEGMENT
    ASSUME CS:CODE,DS:DATA

MAIN PROC
    MOV AX,DATA
    MOV DS,AX

    CALL MULTIPLY
    CALL BIN_OUT

    MOV AH, 4CH
    INT 21H
MAIN ENDP

MULTIPLY PROC
    CALL BIN_IN
```

```
    MOV INA, CX
    CALL BIN_IN
    MOV INB, CX

    MOV AX, INA
    MOV BX, INB
    PUSH AX
    PUSH BX
    XOR DX, DX

    LOOP_:
        TEST BX, 1
        JZ END_IF

        ADD DX, AX

    END_IF:
        SHL AX, 1
        SHR BX, 1

        JNZ LOOP_
        POP AX
        POP BX

        RET
MULTIPLY ENDP

BIN_IN PROC
    XOR BX, BX
    MOV AH, 1
    INT 21H

    INPUT:
        CMP AL, 0DH
        JE EXIT_

        AND AL, 0FH
        SHL BX, 1
        OR BL, AL

        INT 21H
        JMP INPUT

    EXIT_:
        MOV DL, 0DH
        MOV AH, 2
        INT 21H
        MOV DL, 0AH
        INT 21H

        MOV CX, BX
        RET
BIN_IN ENDP

BIN_OUT PROC
    MOV BX, DX

    PRINT:
            RCL BX, 1
            JC PRINT1
            JNC PRINT0

    PRINT1:
        CMP COUNT, 0
        JE END_

        MOV DL, '1'
        MOV AH, 2
        INT 21H
        DEC COUNT
        JMP PRINT

    PRINT0:
        CMP COUNT, 0
        JE END_

        MOV DL, '0'
        MOV AH, 2
        INT 21H
        DEC COUNT
        JMP PRINT

    END_:
        RET
```

```
107
108 BIN_OUT ENDP
109
110 END MAIN
```

Output

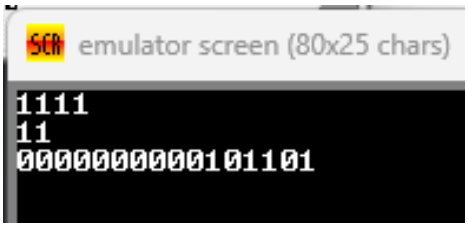


Figure 1: Binary input & output

Problem 6.2

Problem Description

Write a code to take input from user as string and then out the string in reverse. Use stack.

Process

Code:

```
—

DATA SEGMENT

DATA ENDS
CODE SEGMENT
    ASSUME CS:CODE, DS:DATA

START:
    MOV AX, DATA
    MOV DS, AX

    XOR CX, CX
    MOV AH, 1
    INT 21H

INPUT:
    CMP AL, 0DH
    JE EXIT_

    PUSH AX
    INC CX

    INT 21H
    JMP INPUT

EXIT_:
    MOV AH, 2
    MOV DL, 0DH
    INT 21H
    MOV DL, 0AH
    INT 21H

OUTPUT:
    POP DX
    INT 21H
    LOOP OUTPUT

CODE ENDS
END START
```

Output

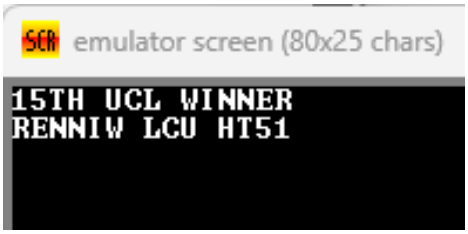


Figure 2: Hexadecimal input & output

Problem 6.2

Problem Description

Write a program that lets the user type some text, consisting of words separated by blanks, ending with a carriage return, and displays the text in same word order as entered, "this is a test" becomes "siht si a tset".

Process

Code:

```
---

DATA SEGMENT
    COUNT DW 0
DATA ENDS

CODE SEGMENT
    ASSUME CS:CODE, DS:DATA

START:
    MOV AX, DATA
    MOV DS, AX

    XOR CX,CX

    MOV AH, 1
    INT 21H

INPUT:
    CMP AL, 0DH
    JE END_

    PUSH AX
    INC CX
    INT 21H
    JMP INPUT

END_:
    MOV DX, 0DH
    MOV AH, 2
    INT 21H
    MOV DX, 0AH
    INT 21H

    MOV BX, SP
    SUB BX, CX
    SUB BX, CX
```

```
36
37      XCHG BX, SP
38      PUSH 20H
39      XCHG BX, SP
40      INC COUNT
41
42  TO_NEW_STK:
43      POP DX
44      XCHG BX, SP
45      PUSH DX
46      XCHG BX, SP
47      INC COUNT
48
49      LOOP TO_NEW_STK
50
51      MOV CX, COUNT
52      MOV COUNT, 0
53      PUSH 20H
54      INC COUNT
55
56  REVERSE:
57      XCHG BX, SP
58      POP DX
59
60      XCHG BX, SP
61      CMP DL, 20H
62
63      JNE SKIP
64
65      MOV AH, 2
66
67  REV_PRINT:
68      POP DX
69      INT 21H
70
71      DEC COUNT
72      JNZ REV_PRINT
73
74      MOV DX, 20H
75
76  SKIP:
77      PUSH DX
78      INC COUNT
79
80      LOOP REVERSE
81
82      MOV AH, 4CH
83      INT 21H
84
85  CODE ENDS
86  END START
```

Output

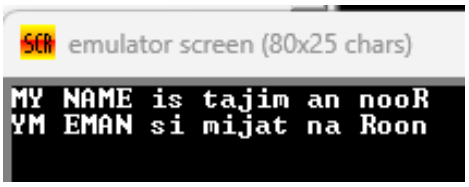


Figure 3: Hexadecimal input & output

Keywords

These are the new keywords learned while solving the Problems

PUSH Used to put a word at the top of the stack.

POP Used to get a word from the top of the stack to the provided location.

XCHG Used to exchange the data from two locations.

CALL Used to call a procedure and save their return address to the stack.

RET Used to return from the procedure to the main program.